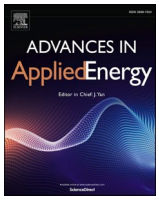
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Variational quantum circuit learning-enabled robust optimization for AI data center energy control and decarbonization

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ABSTRACT

As the demand for artificial intelligence (AI) models and applications continues to grow, data centers that handle AI workloads are experiencing a rise in energy consumption and associated carbon footprint. This work proposes a variational quantum computing-based robust optimization (VQC-RO) framework for control and energy management in large-scale data centers to address the computational challenges and overcome limitations of conventional model-based and model-free strategies. The VQC-RO framework integrates variational quantum circuits (VQCs) with classical optimization to enable efficient and uncertainty-aware control of energy systems in AI data centers. Quantum algorithms executed on noisy intermediate-scale quantum (NISQ) devices are used for value function estimation trained with Q-learning, leading to the formulation of a robust optimization problem with uncertain coefficients. The quantum computing-based robust control strategy is designed to address uncertainties associated with weather conditions and renewable energy generation while optimizing energy consumption in AI data centers. This work also outlines the computational experiments conducted at various AI data center locations in the United States to analyze the reduction in power consumption and carbon emission levels associated with the proposed quantum computing-based robust control framework. This work contributes a novel approach to energy-efficient and sustainable data center operation, promising to reduce carbon emissions and energy consumption in large-scale data centers handling AI workloads by 9.8 % and 12.5 %, respectively.

1. Introduction

Data centers currently account for about 1 % to 1.5 % of global electricity consumption and are responsible for one percent of energy-related greenhouse gas emissions [1]. As the demand for artificial intelligence (AI) models for chatbots and generative applications grows, cyberinfrastructure equipped with vast amounts of computational resources designed for dedicated AI workloads like training and deployment has been leveraged by AI data centers [2]. These comprise of high-performance hardware accelerators like graphics processing units (GPUs), storage systems, servers, and networking architectures. Advances in AI hardware accelerators have enabled energy-efficient computing without compromising processing speed, resulting in a 6 % increase in global data center energy consumption over the last decade despite an explosive growth in computing capacity [3,4]. Even with sophisticated hardware and software designs to handle AI workloads, electricity consumption associated with AI data centers may rise with the increasing popularity of AI-powered applications [5], as exemplified by the growing energy usage of large-scale data centers by 20–40 % annually [1]. Integration of renewable energy sources and battery energy storage to power AI workloads in data centers is a promising strategy to lower reliance on electricity grids while mitigating the carbon footprint and has been adopted by several major providers [1,4]. Apart from the computing hardware, infrastructure components, like cooling and power delivery systems, also contribute significantly to the overall energy consumption [6]. Resource management systems employed to regulate infrastructure operation present an opportunity to establish efficient energy management strategies for AI data centers that incorporate renewable generation and energy storage devices to minimize electricity consumption and accompanying carbon emissions.

Quantum computing holds immense potential across diverse domains by offering exponentially faster computational capabilities [7,8] and potential energy-efficient computing performance [9,10] for various fields such as cryptography [11], computational optimization [12], drug discovery [13], and finance [14]. Quantum computers can offer potential benefits for energy systems optimization problems [15],

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including ones native to data centers. Tasks like optimizing cooling needs, scheduling large workloads, as well as managing energy sources formulated as optimization problems can be tackled with hybrid quantum computing-based solution strategies even at larger scales [16,17]. However, extracting tailored subproblems for decomposition-based optimization and using specific quantum hardware to solve them can pose limitations on the computational advantages derived from such solution strategies. On the other hand, energy management in data centers cast as an optimization problem with varying objectives can be solved with classical optimization solvers [18]. Mathematical programming techniques can also facilitate addressing complexities like renewable energy integration and uncertainties associated with renewable energy generation, weather patterns, and computational workloads [19]. Stochastic optimization strategies have been shown to effectively tackle optimization problems for data center energy management considering such uncertainties [20]. Robust optimization techniques have also been developed to enable energy-efficient and sustainable data center operation by hedging against external disturbances as well as workload uncertainty [21]. It should also be noted that the adoption of sophisticated approaches for temperature and airflow modeling with objectives to minimize metrics like carbon usage effectiveness can induce nonlinearities and nonconvexities in the optimization problem formulation [22]. Although the development of optimization-based strategies has been a promising approach for energy management in data centers, complicated formulations derived by physics-based modeling of energy systems operations and real-world phenomena like dynamic electricity tariffs [23,24], can lead to intractability, especially for data centers that handle large-scale AI workloads.

Optimality of decisions involved in energy-efficient management of data center components can be sacrificed for increased tractability with online control algorithms that attempt to obtain near-optimal controls at each timestep [25]. Although such optimal control methods can incorporate uncertainties like renewable energy generation in the underlying model formulation, which captures workload management and energy storage dynamics in the data center [26], increased complexities can directly affect their computational performance [27,28]. Quantum computing-based approaches to address cooling in building energy systems have also been explored to reduce energy utilization [29]. Optimizing the trade-off between objectives like maximizing the utility of computational services and minimizing costs associated with power consumption can also be realized with control approaches leveraging networks representing the flow of computational tasks and thermal energy in the data center [30]. Reliance on models derived from first principles is a limiting factor of the model-based solution approaches for energy management in data centers, as approximations leveraged to formulate such models may exhibit suboptimal performance in practice [31,32]. Model-free approaches like reinforcement learning can overcome such limitations and have been applied to various tasks in data centers, like workload management [33,34] and controlling cooling systems [35,36]. Deep reinforcement learning, which leverages varying neural network architectures to approximate policies or value functions, has seen increased adoption for thermal energy management in data centers [37]. Cooperative optimization of power consumption required by computing equipment and cooling can be addressed with tailored deep reinforcement learning strategies like multi-agent approaches [38] or integration with rule-based models to manage safety risks [39].

Despite the advantages offered by reinforcement learning approaches for energy management, model-free learning-based approaches applied to data centers are not immune to complexities like sample inefficiency and high computational resource utilization [40], which can be exacerbated by an increase in computation units required to handle large AI workloads [41]. As external disturbances like weather conditions can directly affect the cooling loads while variations in renewable energy generation influence the power consumption levels within the data center, reinforcement learning approaches hedging against uncertainties to derive robust controls can be adopted for energy management [42]. Robust optimization in reinforcement learning can also account for model uncertainties in the system behavior [43] and has been applied to robust control of energy systems [44]. Control techniques integrating deep reinforcement learning with robust optimization rely on classical neural networks for which sample complexities can impair controller performance. Additionally, the use of deep neural networks for function approximation can increase the combinatorial complexity of the surrogate optimization problem [45] and, in turn, increase the computational resource utilization for control computation. Considering the advantages and pitfalls of model-free energy management strategies, it is important to explore the adoption of quantum computing techniques and robust optimization to enable computationally efficient and uncertainty-aware control of energy systems in large-scale AI data centers.

There are several research challenges associated with developing a quantum computing-based robust control framework for energy management in AI data centers. Today's quantum computers, termed as noisy intermediate-scale quantum (NISQ) exhibit limitations in terms of their error rates and size [46,47], which can induce complexities with the application of quantum computing techniques for energy-efficient control. The first challenge lies in adopting quantum algorithms capable of deriving practical utility for real-world problems. Another challenge is to enable a mode of operation that integrates quantum algorithms with classical subroutines, using information from quantum operations to create an optimization problem for control computation in a model-free reinforcement learning setting. The inability to strictly enforce constraints on the various control actuators associated with energy components in the AI data center with a reinforcement learning-based approach [48] can form an additional challenge. A final challenge lies in ensuring that the quantum computing-based reinforcement learning strategy can effectively deal with different uncertainties associated with varying weather conditions and renewable energy generation. Controls obtained with conventional robust optimization strategies can produce conservative solutions [49], resulting in increased data center power consumption. Thus, it is crucial to establish the robustness of the quantum computing-based control strategy without compromising its energy-efficiency capabilities. To address these challenges, we develop a control framework that leverages variational quantum circuits (VQCs) to enable robust optimization in a reinforcement-learning setting for energy management in AI data centers. VQCs are parameterized models executed on circuit-based or gate-based quantum computers [50] and have been extensively used in quantum machine learning algorithms owing to their expressive power on NISQ devices [51]. The control framework termed as VQC-RO leverages a VQC for value function estimation of a state-action pair comprising data center observations, future uncertainty information, and control actions. With efficient value function approximation achieved with a Q-learning based approach, a robust optimization problem can be formulated with a trained VQC, which allows to hedge against future uncertainties like weather conditions or renewable generation and yield robust controls. Classical state-action information is encoded onto the quantum circuits in a way that enables extraction of a linear programming problem with uncertain coefficients in the objective function, which results in an efficient control computation strategy facilitated by a classical optimization solver. We also investigate the power reduction performance and impact of integrating solar generation into AI data centers by conducting multiple computational experiments at various locations in the United States, including California, Texas, and Virginia. Carbon emission levels associated with the data center operations are also analyzed to quantify the impact of the proposed quantum computing-based control framework for efficient energy management in AI data centers. Lastly, the computational efficiency and scalability of the proposed energy management framework are also established with comparisons against classical counterparts. The main contributions of this work can be summarized as follows,

- A novel VQC-RO control framework for energy management in AI data centers that exploits the computational efficiency offered by quantum algorithms executed on NISQ devices along with uncertainty-tackling capabilities of robust optimization.
- Controller design and learning established with a novel mode of operation integrating a variational quantum algorithm for value function estimation with classical optimization to produce robust controls.
- Performance analysis of the proposed control framework in terms of its power consumption and carbon emissions reduction capabilities by computational experiments performed with AI data centers at various locations. A comparative study with the rule-based control method and a deep reinforcement learning approach is also presented.

The remainder of this paper is structured as follows. A digital twin of the AI data center comprising computing infrastructure, battery energy storage, and photovoltaic (PV) generation is constructed in Section 2, along with its Markov decision process formulation. The proposed VQC-RO control framework along with the computational experiments to evaluate its performance efficiency, is presented in Sections 3 and 4, respectively. Finally, conclusions are drawn in Section 5.

2. AI data center digital twin

2.1. Energy modeling

We construct a digital twin of a data center capable of handling intensive computational workloads like AI to enable accurate simulation and visualization of the data center's energy usage, allowing for potential energy efficiency improvements and optimization through high-performance control strategies. In this work, a two-zone data center equipped with HVAC is considered for energy modeling [52]. The AI data center is modeled to include multiple rows of racks, which are standardized enclosures used to mount various computing equipment modules such as servers, switches, storage devices, and networking equipment. We define each rack as a collection of computing units and information technology (IT) equipment fans that draw power to handle corresponding workloads. The instantaneous power consumption associated with the IT equipment in the racks is dependent on the current computing workload, the inlet temperature, and the corresponding power curves [53]. The energy models in [54] are leveraged to prototype the power consumption of rack-mounted servers denoted by E_{rack}. The instantaneous power consumption is computed with the power curve for IT equipment in each rack, which is a piecewise function of the workload ratio. The number of computing units within each rack is chosen such that the power density of a single rack is between 20 and 60 kW to mimic the operation of high-performance computing clusters commonly used for AI training and inference workloads. The IT fans perform active cooling using forced convection followed by dissipation of the heat generated by the computing units into the surrounding environment. The constructed model considers the dynamic interactions between the IT equipment, airflow, and cooling systems in the AI data center to predict energy usage and thermal conditions under different operational scenarios. Another major component of the AI data center is the computer room air handling (CRAH) unit, which removes heat with a chiller system. We equip the server room in the AI data center digital twin with a floor-mounted CRAH unit to facilitate cold air flow through the racks. With the cooling setpoint temperature as input, the power consumption of the cooling system $E_{cooling}$ can be calculated by considering power drawn by cooling components like the cooling tower and the chilled water pump in response to the current room temperature and the ambient air temperature [52].

An overview of the AI data center digital twin with individual components and power flow is shown in Fig. 1. Data centers equipped with battery energy storage systems can contribute to the overall

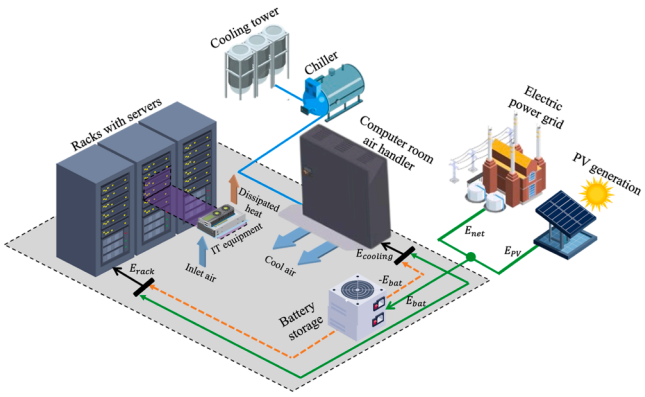


Fig. 1. The AI data center digital twin comprises IT infrastructure, a cooling system, battery energy storage, and PV generation module, depicting the power flow as well as thermal energy flow.

resilience of the electrical grid by not only mitigating the impact of short-term power outages but also providing peak reduction and load balancing capabilities [55,56]. The energy storage system in the digital twin is defined by its capacity C, nominal capacity C_{nom}, charging/discharging efficiency factor ζ, and the capacity loss coefficient η. The nominal capacity determines the amount of power that can be delivered by a fully charged battery under specific conditions, while the capacity loss coefficient measures the degradation of the battery over each charging/discharging cycle. For a rechargeable battery, the nominal capacity and charging efficiency typically vary based on its current state of charge (SOC) and can be represented by piecewise linear functions given by the power capacity curve $C_{nom}(SOC)$ and efficiency curve $\zeta(SOC)$, respectively. The dynamics associated with the battery energy storage system comprising charging and discharging along with the battery capacity degradation can be modeled with Eqs. (1) and (2), respectively. Specifically, Eq. (2) represents the reduction in battery capacity over time, where $|SOC_t - SOC_{t-1}|$ indicates the amount of energy that has been charged or discharged in the previous cycle.

$$SOC_{t+1} = \begin{cases} \min\left\{C_t, SOC_t + E_{bat}\cdot\sqrt{\zeta}\right\} & \text{if } E_{bat} \geq 0 \\ \max\left\{0, SOC_t + E_{bat}\cdot\dfrac{1}{\sqrt{\zeta}}\right\} & \text{if } E_{bat} < 0 \end{cases} \tag{1}$$

$$C_{t+1} = C_t - \eta C_0 \cdot \frac{|SOC_t - SOC_{t-1}|}{2C_t} \tag{2}$$

$$E_{PV} = A_{PV}\eta_{PV}P_f \cdot G_t \tag{3}$$

$$E_{net} = \max\left\{0, E_{rack} + E_{cooling} + E_{bat} - E_{PV}\right\} \tag{4}$$

E_{bat} represents the amount of power drawn by the battery storage for charging and is bounded by its current nominal capacity described by $[-C_{nom}(SOC_t), C_{nom}(SOC_t)]$. In addition to contributing towards efficient energy management, battery energy storage systems also allow easier renewable energy integration. To analyze the impact of adopting renewable energy sources for energy management in data centers, we incorporate photovoltaic (PV) generation to supplement the energy needs of the AI data center and reduce reliance on the electric power grid [57]. In this work, we simulate the installation of PV panels for data center operations, allowing the generated power to be used to minimize energy consumption or charge the data center's battery energy storage system. The amount of energy produced by the PV panels is determined by the area exposed to solar radiation A_{PV}, the efficiency coefficient η_{PV} and the packing factor P_f. The packing factor, or the density of solar cells in a PV module, influences the module's output power and is dependent on the shape of the solar cells used. Using a total solar irradiation of G, the power generated by the PV panels denoted by E_{PV} can be modeled

with Eq. (3). Controlling the cooling setpoint for the CRAH unit in the data center and the charging/discharging levels for the battery energy storage system dictate the electricity consumption incurred by the AI data center. Additionally, power consumption in the data center can be supplemented by PV generation and battery discharging due to the provisions for equipped battery energy storage and renewable generation. The net electric power demand requirement for the AI data center can then be modeled as E_{net} that is drawn from the electric power grid and described in Eq. (4).

2.2. Markov decision process formulation

To enable the development of a model-free approach for efficient energy management in an AI data center, operations encompassing the data center components of computing and cooling as well as energy storage are modeled as a Markov decision process (MDP). An MDP is defined by the state space S, the action space A, the reward function r, a discount factor γ, and the transition dynamics. At time t, we consider the state vector $s_t \equiv (x_t, w_t)$ to include the measurable states of the AI data center x_t and future uncertainty information w_t. Environmental parameters like the air temperature of the server room and ambient air temperature of the external surroundings, along with the state of charge of the energy storage device, constitute the data center state x_t. As the computational workload directly contributes to the power consumption levels in the AI data center, the workload ratio is also considered as a state variable. Enabling time awareness while addressing sequential decision-making problems involving state variables with seasonal variations is important to improve the learning capabilities of the reinforcement learning agent [58]. To accommodate time awareness in the reinforcement learning agent, cyclical feature encodings using sine and cosine transformation for the day of the week and month are included in the data center state x_t spanning the state space $s_t \in \mathbb{R}^8$. Uncertainties like future forecasts for ambient temperature, carbon intensity levels, or PV generation can be included in the uncertainty vector w_t. The state variables can be adjusted based on specific objectives. For a sequential decision-making problem tasked with minimizing net carbon emissions, it is important to consider the current carbon intensity as a state variable in x_t and forecast carbon intensity as w_t to form the state space $s_t \in \mathbb{R}^9$. As the constructed digital twin for the AI data center allows provisions for controlling the cooling setpoints and energy storage charging and discharging levels, the action space can be described as $a_t \in \mathbb{R}^2$. The action space can be limited to prevent instabilities during training and mitigate extensive computational resource utilization. The action variable corresponding to the cooling setpoint is then reformulated as the change of the previous cooling setpoint constrained in $[-\delta T_{max}, \delta T]$. Similarly, the charging/discharging actions for the battery energy storage system are restricted within $[-C_{nom}(SOC_t), C_{nom}(SOC_t)]$. Using the known bounds on the control actions, we further define the action space as $a_t \in [-1,1]^2$. The transition dynamics for the MDP are deterministic and align with the energy model constructed for the AI data center digital twin. The reward structure for the MDP can be modified depending on the reinforcement learning task. For power consumption reduction and carbon emission reduction, the reward function at any time t can be defined as $r_t = -E_{net,t}$ and $r_t = -c_t E_{net,t}$, respectively, where c_t and $E_{net,t}$ indicate the carbon intensity and net power drawn by the AI data center in response to its current state and chosen control actions. As the objective of reinforcement learning agents is to identify a policy $\pi : S \rightarrow A$ which maximizes the expected discounted rewards over an infinite horizon $E_{a\sim\pi}\left[\sum_{t=0}^{\infty}\gamma^t r_t\right]$, the reward function is scaled as $(R_{min} - r_t)/R_{min}$ to bound the individual and total rewards.

3. VQC-RO control framework

This section presents the quantum computing-based robust control framework for energy management in AI data centers. The proposed methodology employs a Q-learning approach to estimate the state-action value function using a VQC, which represents the cumulative sum of discounted rewards over an infinite horizon. Encoding the input data onto the quantum circuit, applying parametric gate operations, and measuring quantum states involved in the VQC are the key steps comprising value function estimation. Extracting the robust optimization problem that incorporates future uncertainties from the trained VQC is solved for computing controls at each timestep. The training process for the VQC involves using a Q-learning based approach to update the gate parameters and estimate the target function, while the implementation phase involves using the trained VQC to obtain control actions for the AI data center by solving a robust optimization problem. An overview of the VQC-RO control framework for energy management in AI data centers is also shown in Fig. 2.

3.1. Variational quantum circuit-enabled robust optimization

To address the problem of energy management in AI data centers modeled as a MDP, a Q-learning approach is employed to estimate the state-action value. Q-learning aims to obtain a value function $Q(s,a)$ that satisfies the Bellman equation in Eq. (5), wherein $\pi(s)$ represents a deterministic policy function. Here, we leverage a VQC to parameterize the state-action value or the Q-value function capable of estimating the cumulative sum of discounted rewards over an infinite horizon with the control trajectory initialized with the (s,a) pair. Executing a VQC consists of three key steps: encoding the input data, applying the parametric gate operations or the variational circuit, and measuring the quantum states. Translating a classical vector into a quantum state is referred to as quantum embedding or encoding. To encode the state action pair (s,a) comprising of $s \equiv (x,\ w)$, separate quantum embedding schemes are employed to ease formulation of the optimization problem for control computation in the subsequent steps. The observation vector $x \in \mathbb{R}^8$ is encoded with the amplitude encoding scheme with a four-qubit quantum state. A quantum state $|\psi\rangle = \sum_{i=1}^{8}\frac{x_i}{||x||_2}|i\rangle$ can be prepared by normalizing the data center observations with the amplitudes of the three-qubit computational basis states. Loss of information by using one fewer degree of freedom can be prevented by encoding the norm of the observation with a parametric quantum gate. The Rx quantum gate performs a specified rotation about the x-axis of a single-qubit state. By establishing bounds for the amount of rotation, the state vector norm can be mapped to θ_x. The input state $|\psi_x\rangle$ generated for the observation x can then be described by Eq. (6). A similar transformation is applied to encode the control variable $a \in [-1,\ 1]$ and the uncertainty w by rotation about the x-axis with Rx gates and specific angles θ_a and θ_w obtained from known bounds. Two single-qubit quantum states $|u\rangle = Rx(\theta_a)$ with $\theta_a = \pi \cdot a$ and $|w\rangle = Rx(\theta_w)$ are prepared by encoding the classical information. It should be noted that for state-action value estimation, the parameters in the data encoding step $(\theta_x, \theta_a, \theta_w)$ remain fixed for each classical state and action pair.

$$Q(s,a) = E_{s'}[r + \gamma Q(s', \pi(s'))] \tag{5}$$

$$|\psi_x\rangle = |\psi\rangle \otimes Rx(\theta_x)|0\rangle = |\psi\rangle \otimes \begin{pmatrix} \cos\frac{\theta_x}{2} \\ -i\sin\frac{\theta_x}{2} \end{pmatrix} \tag{6}$$

$$A_i = \left(\alpha_{u,i}\cos\theta_u/2 + \beta_{u,i}\sin\theta_u/2\right)\left(\alpha_{w,i}\cos\theta_w/2 + \beta_{w,i}\sin\theta_w/2\right) \tag{7}$$

$$Q(s,a;\Theta) = Q_w \cdot m + Q_b \tag{8}$$

After the initial state following data encoding is prepared, a VQC with adjustable parameters is used to process the input quantum state. As shown in Fig. 3, pooling gate operations [59] are also applied between consecutive qubits comprising the quantum state $|\psi_x\rangle$ which

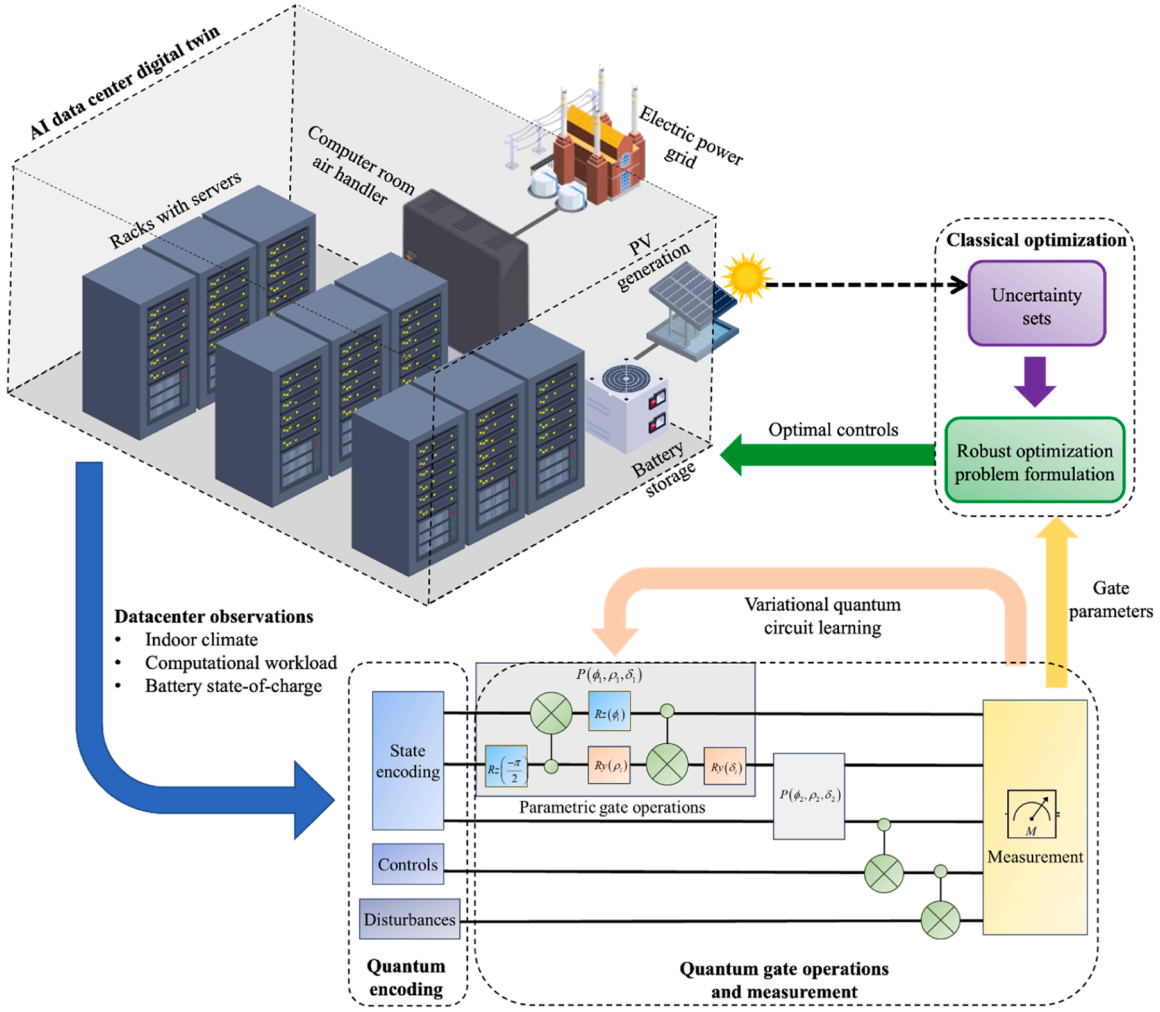


Fig. 2. An overview of the energy management in an AI data center by computing controls with the proposed hybrid variational quantum circuit enabled robust optimization framework.

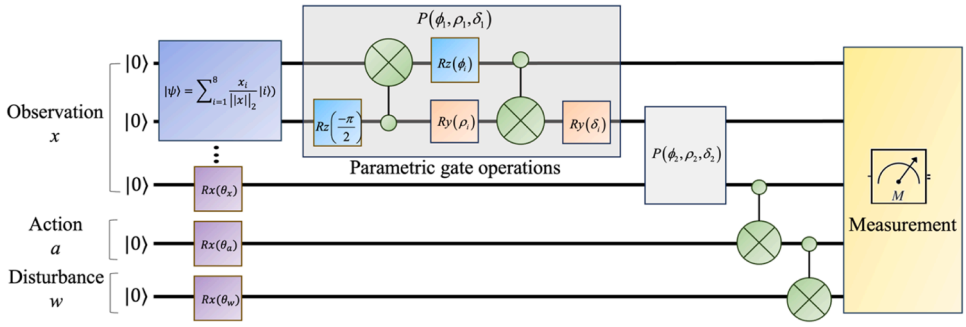


Fig. 3. Quantum ansatz for the variational quantum circuit comprising data encoding steps, parametric gate operations, and the measurement operator.

represents the data center observations. The pooling operator $P(\phi_i, \rho_i, \delta_i)$ involves a series of gate operations between the i'th source and sink qubit pair, including a fixed inversion of the sink qubit along the z-axis and an inverted controlled-NOT operation. These are followed by rotation of the source and sink qubits along the z-axis and y-axis by ϕ_i and ρ_i angles with the Rz and Ry gates, respectively. Finally, a controlled-NOT operation followed by a rotation of the sink qubit state by a parameter δ_i along the y-axis is applied. These pooling operators are applied to each qubit pair comprising the four-qubit quantum state $|\psi_x\rangle$ sequentially, to enhance the entangling capability of the VQC and distill the information within the quantum state into a single-qubit sink qubit. This qubit state is further entangled with the single-qubit states $|u\rangle$ and $|w\rangle$ using controlled-NOT operations, which directly contribute to the efficiency of variational quantum algorithms [60]. Following the parameterized unitary gate operations in the VQC, the resulting n-qubit quantum state can be represented by 2^n quantum amplitudes corresponding to the individual basis states. The i'th quantum amplitude can be represented by A_i as shown in Eq. (7) where $\alpha, \beta \in \mathbb{C}$ that satisfy the complementary condition $\alpha \cdot \beta = 0$. Measuring this quantum state yields probabilities of the quantum state collapsing into the i'th computational basis state as

$m_i = |A_i|^2$. The measurement vector m is scaled by parameters Q_w and Q_b as shown in Eq. (8) to estimate the state-action value function for the input (s, a).

$$RO(\mathrm{x}, \Theta): \max_{\tilde{\mathrm{u}}} \min_{\mathrm{w}} \sum_{i=1}^{2^{\mathrm{n}}} \mathrm{c}_{\mathrm{i}}^2 \cdot \mathrm{f}(\mathrm{w}) \tilde{\mathrm{u}} \tag{9}$$

$$\text{s.t. } \mathrm{c}_{\mathrm{i}} = \mathrm{Q}_{\mathrm{w}} \cdot \lambda_{\mathrm{i}}(\mathrm{x}, \Theta) \ \forall \mathrm{i} \in [1, 2^{\mathrm{n}}] \tag{10}$$

$$0 \leq \tilde{\mathrm{u}} \leq 1 \tag{11}$$

$$\mathrm{f}(\mathrm{w}) \in \mathrm{U} \tag{12}$$

Using a trained VQC with parameters Θ to estimate the value for a given state-action pair, a robust optimization problem can be formulated to extract the controls with a deterministic policy $\pi(x)$. The VQC is trained to estimate the cumulative discounted rewards with the current data center observations and actions along with perfect information of future uncertainty. At any time step t, the max-min problem in Eqs. (9)–(12) with the current set of VQC parameters and data center observations x_t is solved to compute auxiliary variables $\tilde{u}$. The robust optimization formulation hedges against future uncertainties w by incorporating uncertainty sets for the function $f(w) = \cos^2\theta_w/2$ in Eq. (9). The uncertainty sets for variables w can be further transformed using the same transformation scheme and known bounds to derive uncertainty sets U in Eq. (12). It should also be noted that the auxiliary variable $\tilde{u}$ is chosen such that the resulting optimization problem is linear for faster solution times. The objective function in Eq. (9) is to maximize the worst-case realization of the scaled measurement values which represent the state-action value. The coefficients of this linear objective comprise the transformed uncertain variables as well as squares of linear combinations of encoded observations represented by $\lambda_i(x, \Theta)$. These linear transformations are a direct result of the unitary gate operations applied to the input quantum state that encodes the data center observations x_t. Constraints like Eq. (11) can also be leveraged in the robust optimization problem formulation to enforce solution search within the feasible action space. Following the solution procedure, the optimal values of the auxiliary variables can be further used to compute optimal control actions with $\pi(x_t) \equiv G(\tilde{u}^* = argmax_{\tilde{u}} RO(x_t, \Theta))$ where the mapping G is obtained by inverting the transformation obtained from Eqs. (7) and (8). This mapping is used to compute control actions for the MDP at any time t and can be described as $a_t = G(\tilde{u}) = 1/\pi \cos^{-1}(\sqrt{\tilde{u}})$.

3.2. Controller learning and implementation

Learning in a VQC involves measurement of the quantum state obtained after parametric gate operations to estimate the target function and updating the gate parameters by adhering to the minimization of a specified loss function. Here, as the VQC approximates the state-action value function, the VQC is trained with a Q-learning based approach, as mentioned in Section 3.1. The VQC is initialized with parameters Θ sampled from a uniform distribution $U(-\pi, \pi)$. A replay buffer D is used to store transitions and serves as experience replay to help increase sample efficiency and prevent training instabilities [61]. During the training process, each transition at timestep t includes the current state s_t comprising data center observations x_t and perfect knowledge of the future uncertainty w_t, applied control actions a_t, the obtained reward r_t, and the following state s_{t+1}. To balance exploration and exploitation while training the hybrid VQC-RO controller, an epsilon-greedy strategy is used. This strategy determines the probability of choosing controls that maximize the value estimate, as shown in Eq. (13). During the exploration phase, instead of randomly sampling controls, a rule-based controller (RBC) can be used to ensure reasonable charging and discharging levels of energy storage devices equipped in the AI data center. Although typical Q-learning schemes rely on random exploration, the use of a rule-based control can help to avoid unnecessary exploration and subsequent computation. The probability is decayed by a factor of ϵ_{decay} with each timestep as the training progresses. It should also be noted that during the training phase of the controller, $RO(x_t, \Theta)$ is a deterministic optimization problem due to perfect information availability of the future uncertainties and can be denoted by $DO(x_t, \Theta) \equiv \max_{\tilde{u}} \left\{ \sum_{i=1}^{2^n} c_i^2 \cdot f(w) \tilde{u} \right\}$ subject to constraints in Eqs. (10) and (11). A batch of transitions of size B_{size} is sampled from the replay buffer D. As the objective of Q-learning is to approximate the Bellman's equation, the targets for all state-action pairs are computed by maximizing the value function over the set of next states with $y_t = r_t + \gamma \cdot max_{\tilde{u}} DO(x_{t+1}, \Theta)$. It should be noted that computing the targets y_t requires solving the deterministic optimization problem instead of the robust counterpart owing to the perfect knowledge of the future states within recorded trajectories. Computing these target values requires solving an optimization problem with a classical solver for each transition in the sampled batch.

$$P(a_t = G(argmax_{\tilde{u}} RO(x_t, \Theta))) = 1 - \epsilon \tag{13}$$

$$\frac{\delta \mathrm{Q}}{\delta \Theta} = 2\mathrm{Q}_{\mathrm{w}} \cdot \left\{ \mathrm{Q}\left(\mathrm{s_t}, \mathrm{a_t}; \Theta + \frac{\pi}{2}\right) - \mathrm{Q}\left(\mathrm{s_t}, \mathrm{a_t}; \Theta - \frac{\pi}{2}\right) \right\} \tag{14}$$

Using the computed targets, the VQC-based value function estimator can be updated in a supervised manner to minimize the mean squared error objective. Gradients of the objective with respect to individual parameters of the VQC are needed for a gradient descent step. We use a parameter shift rule to estimate the gradients of the VQC-based value function approximator, as given by Eqs. (8) and (14). Eq. (14) requires the computation of value function estimates with two sets of gate parameters for obtaining gradients of the loss function. Finally, the scaling parameters Q_w and Q_b are updated with a gradient descent step. These steps are repeated over multiple training episodes for circuit learning during the training phase of the hybrid VQC-RO controller. During the implementation or the evaluation phase, only data center observations x_t are available at any time step. To hedge against future uncertainties w_t, the robust optimization problem $RO(x_t, \Theta)$ is solved with formulated uncertainty sets and a current set of gate parameters Θ to obtain control actions a_t. During the implementation stage, the VQC parameters can also be updated at larger intervals with the training procedure described above to ensure the adaptability of the controller to previously unseen data center trajectories. An overview of the controller learning procedure is also presented in Fig. 4.

4. Computational experiments for performance evaluation

We conducted several computational experiments with the AI data center digital twin to evaluate the performance of the proposed VQC-RO control framework. For our experiments, the data center digital twin is set up to mimic large-scale AI workloads by adjusting the power density levels of each rack in the data center. Data center operations at three locations in the United States are simulated to analyze the energy management performance of the VQC-RO control strategy in varying weather conditions. The battery energy storage system in the AI data center is sized such that it can supplement its net power consumption requirements. Location-specific simulations are performed with historical weather and carbon intensity data recorded in 2021–2022 at three locations in Los Angeles (California), Dallas (Texas), and Ashburn (Virginia). The carbon intensity levels indicate the amount of carbon emissions released by the grid for generating one unit of electricity. The higher concentration of data centers at these locations in the United States as compared to other locations forms the basis for evaluation in our experiments. We use real-world computational workload data from Alibaba for each AI data center under consideration [62].

Apart from the VQC-RO control framework, a rule-based controller

Algorithm 1 VQC-RO controller learning

Initialize the VQC circuit with parameters $\Theta \equiv (\phi_i, \rho_i, \delta_i)$.
Initialize an empty replay buffer $\mathcal{D}$ and hyperparameters like batch size B_{size}.
for each episode **do**
  for $t = 1$: episode length **do**
    Observe data center state $s_t = (x_t, w_t)$
    Compute action a_t with epsilon-greedy strategy.
      Sample $v_r \sim U(0,1)$
      $$a_t = \begin{cases} RBC(x_t) & if\ v_r < \epsilon \\ Solve\ RO_{\widetilde{u}}(x_t, \Theta) & if\ v_r \geq \epsilon \end{cases}$$
    Apply controls and observe new state s_{t+1}, reward r_t
    Store (s_t, a_t, r_t, s_{t+1}) in replay buffer $\mathcal{D}$
    if $|D| \geq B_{size}$ **then**
      Randomly sample a batch of transitions, $B = (s, a, r, s')$ from $\mathcal{D}$
      Solve optimization problem $DO(x', \Theta)$ to compute targets:
      $$Q_y(s', a'; \Theta) \rightarrow \max_{\widetilde{u}} DO(x', \Theta)$$
      $$y = r_t + \gamma \cdot Q_y(s', a'; \Theta)$$
      Compute loss function: $L(\Theta) = E_{(s,a,r,s')}[(Q(s, a; \Theta) - y)^2]$
      Estimate gradients $\frac{\delta L}{\delta \Theta}$ by computing $\frac{\delta Q}{\delta \Theta}$ with parameter shift rule.
      Update circuit parameters Θ with gradient descent.
    end if
    $\epsilon = \epsilon \cdot \epsilon_{decay}$
  end for
end for

Fig. 4. Algorithm for training and implementing the proposed VQC-RO controller for energy management in the AI data center.

(RBC) and the deep deterministic policy gradient (DDPG) method are applied for energy management in these AI data centers for benchmarking purposes. The RBC controller relies solely on the proportional-integral-derivative loop to adjust controls based on the errors associated with the measured computing room temperature and the cooling setpoint. A model-free deep reinforcement learning method, DDPG, is chosen to benchmark the VQC-RO control framework, which adopts a reinforcement learning approach to train the VQC-based value function estimator. As the proposed control framework does not rely on a model derived from first principles, model-based robust optimization techniques [63,64] are not selected as baselines to ensure a fair comparison. The model-free approaches can also aid in evaluating the adaptability of the controllers under varying AI data center characteristics. For the reinforcement learning-based approaches, a reward function to minimize the net power consumption of the data center is utilized along with a discount factor $\gamma = 0.995$. The hyperparameters for both these approaches are also kept consistent for a fair comparison. It should also be noted that the training process for the learning-based controllers is conducted with historical data in 2021, while their performance is evaluated with recorded weather conditions in 2022. Control computation for the VQC-RO control framework leverages box uncertainty sets during the evaluation process. Initially, we apply the energy management strategies to the AI data centers at various locations without considering the PV integration to analyze their impact on existing data centers. The adaptability capabilities of the control strategies are also further evaluated by introducing PV generation capabilities at each AI data center to measure the power consumption reduction performance in response to cyclical uncertainties of solar radiation levels. Finally, the net carbon emissions levels associated with electricity consumption are also analyzed to estimate the reduction in carbon dioxide emissions with advanced energy management strategies in the AI data centers.

4.1. Reduction in power consumption

Data center operations at all three locations are simulated over the year 2022 with different energy management strategies. The annual net power consumption by each AI data center without considering PV generation is depicted in Fig. 5. The highest power consumption levels are observed in Texas, which can be attributed to the large amounts of power drawn for cooling the data center with relatively higher ambient air temperatures. On the other hand, the least power consumption is incurred for the AI data center located in Virginia as a result of colder temperatures during the winter months. The annual power consumption obtained with the proposed VQC-RO control strategy is significantly lower than the DDPG controller for AI data centers at all three locations, with the percentage reduction ranging between 7.3 to 12.5 %. For the states of California and Virginia, VQC-RO demonstrates a reduction of approximately 5 % in its power consumption levels as compared to the

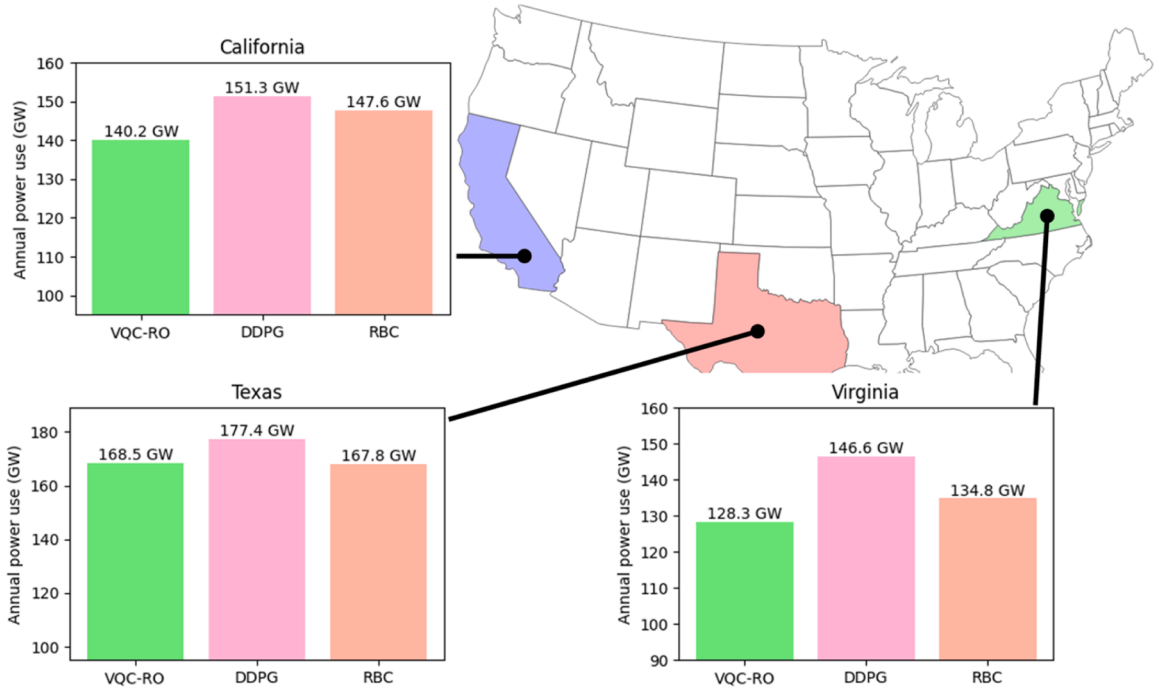


Fig. 5. The annual power consumption of an AI data center observed with various energy management strategies in different locations of the United States.

RBC strategy, while only a 0.4 % variation is observed in the state of Texas.

To analyze the seasonal variations in power consumption, we also measure the net monthly consumption for months representative of the seasons of winter, spring, summer, and fall, as reported in Table 1. It can be observed that the VQC-RO consistently produces lower power consumption across all seasons in California and Virginia. For the colder months in Texas, the RBC strategy exhibits improved performance over both learning-based energy management techniques. The lower energy consumption in January achieved with RBC also aligns with the annual power consumption in Texas, as shown in Fig. 5. Additionally, the DDPG energy management strategy performs worse than the RBC or VQC-RO in each season at all AI data center locations, with only two exceptions and limited variability. We also plot the weekly trend for each month and location obtained with the various control techniques in Fig. 6. From these trend lines, sudden drops or rises in power consumption levels are observed in the Texas data center in the months of January and April. These abrupt changes could be attributed to extreme weather events like the North Texas snowstorm of 2022 and are responsible for varying power consumption reduction performance in contrast to remaining seasons where only slight fluctuations in trends are observed.

4.2. Renewable energy integration and carbon emission reduction

To further validate the robustness of the applied controls, we conduct experiments by integrating PV generation into the data center operations as described in the digital twin section. The learning-based control techniques are also updated to include PV generation levels as a state variable. It should be noted that for the proposed VQC-RO strategy, the qubit count remains fixed despite an increase in dimensionality owing to the logarithmic scaling of the amplitude encoding scheme used to encode data center observations. However, the uncertainty variable is updated to consider solar radiation levels instead of ambient temperature. To visualize the effectiveness of power consumption reduction by considering solar radiation uncertainties during the day, we plot various consumption levels for the AI data center situated in Virginia. For this AI data center, power consumed by computing infrastructure and cooling systems, along with the net power drawn from the electric grid, are plotted for one-week periods of January and July. These periods are chosen due to the distinction in temperature and solar radiation levels in Virginia. As seen from these plots, the key contributor to the varying power consumption levels is the cooling load. Apart from the reduction in power consumption, the integration of PV into the AI data center and consideration of solar radiation uncertainties with the proposed VQC-RO control framework resulted in significant carbon dioxide emission reduction, as shown in Fig. 7. Renewable integration combined with applying robust controls obtained with the VQC-RO strategy results in an annual carbon emission reduction of 6.9 %, with only 3.49 % in winter and as high as 9.85 % in the summer season. Summers in Virginia exhibit a significant drop in net power consumption of the AI data center owing to the consistent PV generation. As the net electricity consumption can be supplanted with PV, small reductions in carbon emissions are observed in winter owing to the

Table 1
Monthly power consumption of the AI data center (in GW) situated at different locations to meet the computing and HVAC load demand obtained with various control techniques.

Month	California			Texas			Virginia		
	RBC	*DDPG*	*VQC-RO*	*RBC*	*DDPG*	*VQC-RO*	*RBC*	*DDPG*	*VQC-RO*
January	9.8	11.8	9.3	11.3	13.5	13.4	8.7	10.8	8.2
April	10.2	11.7	9.9	16.1	15.7	15.9	9.1	11.1	8.8
July	12.0	11.4	11.4	16.9	16.4	16.3	15.5	14.9	15.5
October	16.4	14.8	15.3	10.8	12.6	9.6	10.8	11.2	9.5

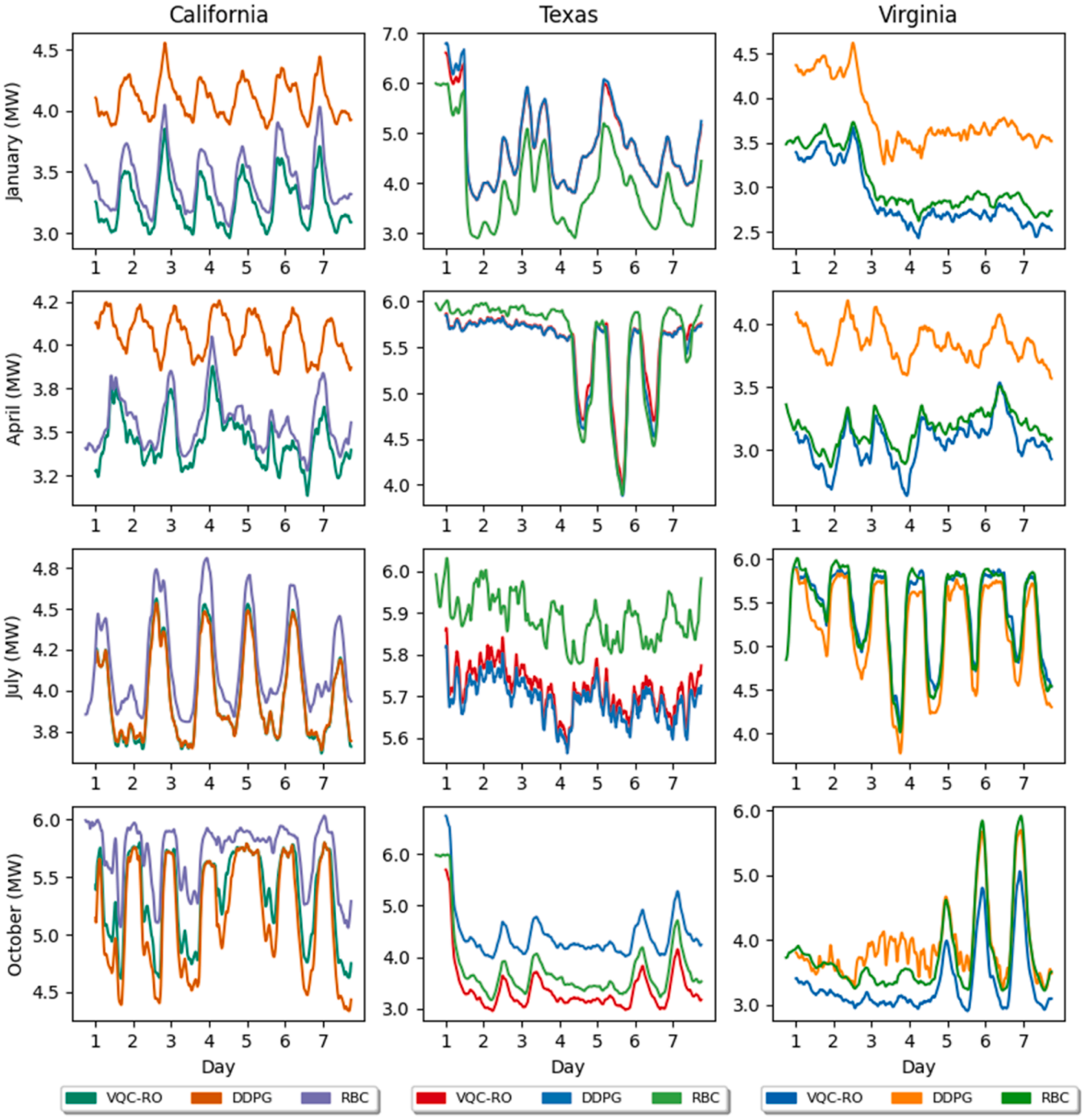


Fig. 6. Net power consumption of the AI data centers situated at three locations with varying weather patterns incurred with the proposed VQC-RO control framework and the baseline methods.

limited PV generation in the winter in contrast to the summer season as seen in Fig. 7. It should also be noted that the average carbon emission level attained in summer is 1819 tonnes.CO_2 and is much higher than that of the emission levels in winter with 1227 tonnes.CO_2. This is mainly because energy requirements associated with cooling operations in AI data centers remain high in summer, accompanied by significant carbon emissions despite the high PV generation in the season. On the other hand, the RBC and the DDPG controllers yield a 4.6 % and 3.5 % increase over the VQC-RO controller, respectively, in terms of yearly net carbon emission levels for the AI data center situated in Virginia.

4.3. Computational efficiency

In addition to demonstrating the energy efficiency and adaptability of the proposed VQC-RO control framework for energy management in AI data centers, we also quantify the utility of the VQC used within the framework. For comparison with the classical counterparts, a neural network is used as the state-action value function approximator. To accommodate the robust optimization problem formulation in the subsequent steps of the proposed control framework, a feedforward neural network with rectified linear units as the nonlinear activation is used [44,45]. We conduct multiple computational experiments with various numbers of hidden layers in the feedforward neural network to analyze

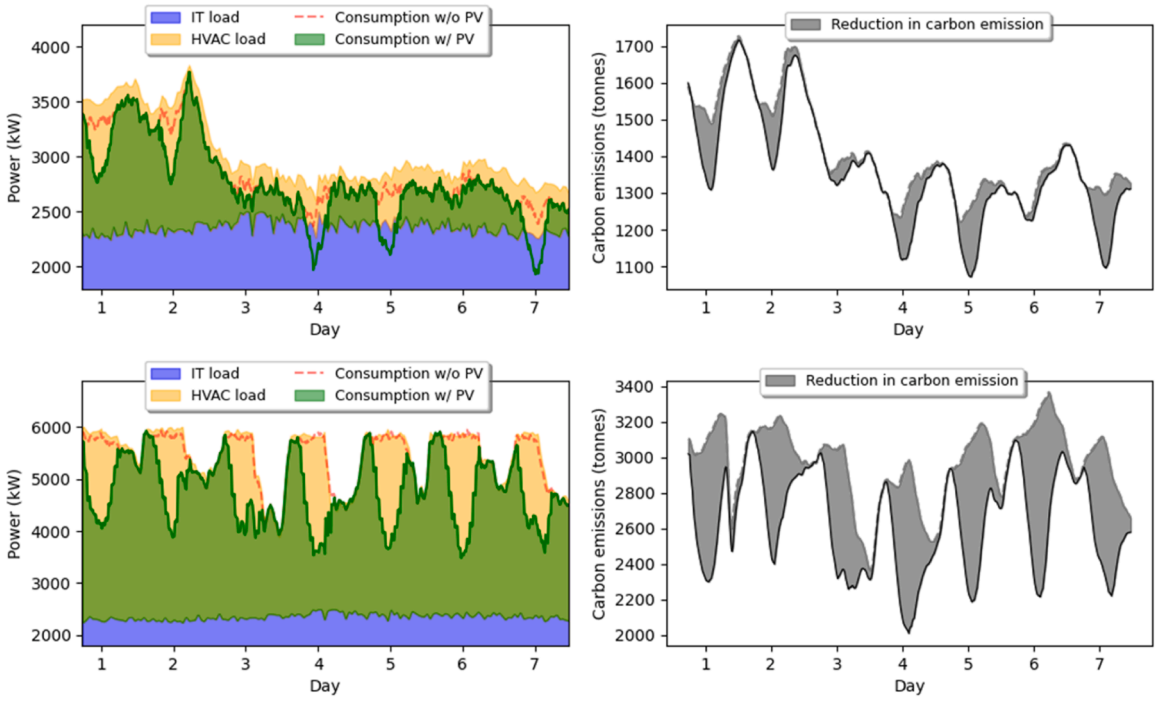


Fig. 7. The effect of PV integration into the AI data center in Virginia demonstrated by the net incurred load demands and the reduction in net carbon emissions in the months of winter (January) and summer (July).

its effect on the control computation task. Similar experiments are conducted by repeating the selected quantum ansatz described in Section 3.2. The number of parameters comprising each combination are presented in Fig. 8. As evident from this figure, the number of parameters in the classical neural network scale exhibits a significant growth with layer repetitions compared to the quantum circuit. The number of training parameters in a classical feedforward neural network used as a value function approximator scales as $O(r{\cdot}h{\cdot}(|S|+|A|))$ where h represents the number of hidden units with the state space and action space of the underlying MDP denoted with S and A, respectively. In contrast, the number of training parameters comprising the VQC increases linearly with the number of ansatz layer repetitions and can be denoted by $O(r)$. Here, r represents the number of layer repetitions for both VQC and the feedforward neural network.

As demonstrated by the energy management performance of the VQC-RO control framework in Sections 4.1 and 4.2, the VQC is able to efficiently approximate the state-action value function with fewer parameters. As the number of parameters influences the learning capabilities and sample complexity associated with parametric models, classical neural networks can be expected to utilize an increased amount of computational resources during training as compared to their quantum counterparts, owing to the high expressivity of the VQC [51]. Furthermore, the choice of value function estimator directly affects the computational resource utilization required for computing controls with the classical optimizer. A feedforward neural network results in a mixed-integer linear program as the surrogate model, while the VQC as a function approximator, yields a linear program. The number of variables and constraints comprising the robust optimization problem formulation in the proposed control framework grows quickly with the number of layer repetitions and hidden units in the feedforward neural network. On the other hand, the number of variables and constraints in the optimization problem formulated with the VQC remain fixed and are not dependent on the ansatz repetitions. This can be substantiated by the solution times observed with classical neural networks and the quantum circuits, as presented in Fig. 8. Even though the ansatz repetitions are accompanied by an increase in gate parameters of the VQC, the time

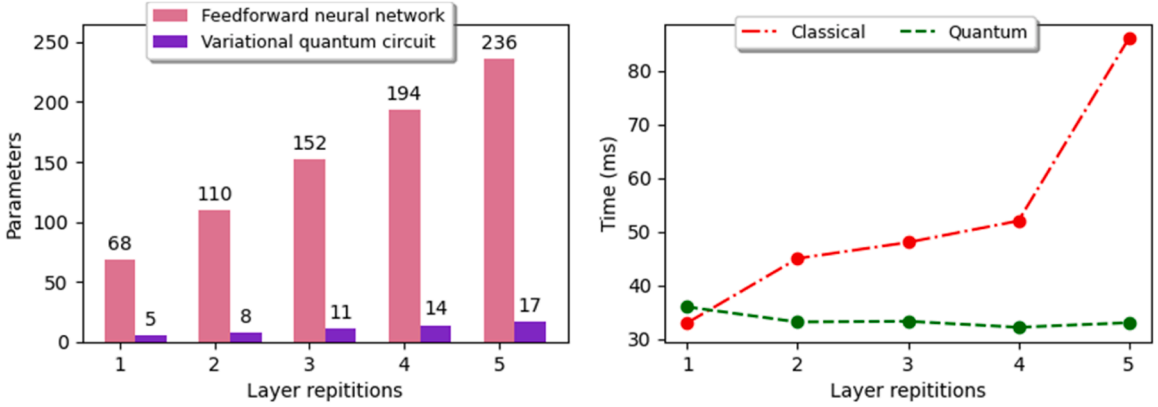


Fig. 8. Comparison between the variational quantum circuit within the proposed VQC-RO control framework and its classical counterparts in terms of the number of trainable parameters and computational resource utilization for control computation.

required to compute controls with the resulting robust optimization problem is consistent over all experiments. The use of a quantum circuit within the proposed VQC-RO control framework not only exhibits scalability in terms of its learning capabilities but also demonstrates computational efficiency for control computation with the subsequent optimization problem solving by a classical optimizer.

5. Conclusion

In this work, a control strategy leveraging quantum computers and robust optimization was developed for energy management in data centers capable of handling AI workloads. By integrating VQCs with robust optimization, the proposed control framework addressed the challenges of producing robust controls by hedging against weather and renewable generation uncertainties while ensuring computational efficiencies offered by quantum algorithms. The computational experiments conducted at various locations in the United States have demonstrated the framework's ability to significantly reduce power consumption and carbon emissions associated with data center operations. Additionally, the comparative study with other methods has confirmed the superiority of the VQC-RO framework. The proposed approach for energy management in AI data centers derived practical utility from NISQ devices while producing competitive results with a power consumption reduction of over 12.5 % and carbon emissions reduction of 9.8 % compared to conventional control techniques and classical computing counterparts. The use of VQC as a value function approximator in the proposed control framework for energy management in AI data centers benefits from its expressivity while facilitating a computationally efficient and scalable approach for control computation in contrast to its classical counterparts.

CRediT authorship contribution statement

Akshay Ajagekar: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Fengqi You:** Writing – review & editing, Visualization, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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